

Label-efficient emulation of conformal reference sets for wearable activity recognition

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Abstract

Labeling every window in a calibration pool can be costly for wearable activity recognition. We propose joint rank calibration (JRC) to reduce this effort while preserving the prediction sets defined by the complete pool. A coupled rank-state recursion calibrates both quantile boundaries across the declared query checkpoints. Labels are acquired in uniform random order without replacement, and acquisition stops when the endpoint prediction sets differ by at most a specified mean number of labels on an unlabeled monitoring batch. The resulting confidence event guarantees reference-set containment for every input and controls average set inflation on that batch. The main evaluation uses seven public datasets and 354 classifier cases, including a separately locked confirmation. At confidence error 0.05 and tolerance 0.5, JRC saves 23.67% of reference labels on the confirmation dataset, compared with 17.91% for Bonferroni hypergeometric intervals and 11.74% for the primary uniform-prior confidence sequence. A subsequent prior diagnostic improves that baseline to 15.33%. An additional, separately specified nine-user confirmation contributes 45 cases and yields 34.03% savings for JRC versus 25.13% for the stronger-prior baseline. The results identify when joint rank calibration provides useful savings and how the benefit depends on reference-set quality.

Keywords: Human activity recognition, Conformal prediction, Label acquisition, Finite-population inference, Sequential calibration

1 Introduction

Wearable human activity recognition (HAR) systems often encounter users whose movements differ from the training cohort. Under this shift, predicted class probabilities can become poorly calibrated [1, 2]. Prediction sets represent uncertainty by retaining several plausible activities, but constructing a calibration reference requires labeled sensor windows. Reducing this labeling effort is especially useful when a reference pool contains many repeated measurements from several users.

Conformal prediction constructs prediction sets under specified calibration assumptions [3, 4]. Hierarchical and beyond-exchangeability methods address dependencies and distribution shifts through additional structure [5, 6]. We focus on a complementary acquisition problem: given a frozen classifier and a fixed finite pool, how many labels are needed to preserve the prediction sets defined by labeling the whole pool? We seek an output that contains those reference sets while allowing a controlled increase in their average size on an unlabeled monitoring batch.

We propose joint rank calibration (JRC) for this purpose. Labels are queried in uniformly random order without replacement. A coupled state records how many sampled ranks lie below the reference rank and whether the reference item has appeared. Its recursion computes the probability that either quantile boundary is crossed at any declared checkpoint. This joint calculation allocates the confidence budget across the full rank path. Acquisition stops when the prediction sets at the two running endpoints are sufficiently close on the monitoring batch, and the upper endpoint defines the returned predictor.

Sequential calibration and prediction-set-based stopping build on established work. Ilyas et al. study labeling and set-size costs under independent sampling [7], while confidence sequences support inference at data-dependent stopping times [8–10]. JRC uses these foundations to control preservation of a fixed finite reference. Its specific contribution is the coupled rank-path calibration and its integration with a stopping rule expressed directly in prediction-set size.

The study makes three contributions:

1. A finite-reference acquisition formulation that connects high-probability set containment with an interpretable mean set-inflation tolerance.
2. A joint rank-state recursion that calibrates both quantile boundaries over the declared checkpoints and supports label-free evaluation of the stopping criterion.
3. An evaluation of 354 classifier cases on seven public datasets, followed by 45 additional cases in a separately specified confirmation cohort. Comparisons examine interval methods, acquisition controls, mechanism ablations and stronger confidence-sequence priors.

The classifiers are logistic regression (LR), MiniRocket (MR) and an InceptionTime-style (IT) network. Six datasets support development, USC-HAD provides the main confirmation, and the additional cohort tests the unchanged method. The analysis relates label savings to reference-set quality and computational cost.

2 Related work

2.1 Wearable recognition and uncertainty

Recent HAR frameworks improve representation learning and adaptation. STF-AL combines temporal, spatial and frequency representations with adversarial domain adaptation [11]. CCT-HAR uses cross-modal sample mining and temporal alignment [12], and TSDFSA studies attention-based feature selection for multiple sensors [13]. These methods motivate controlled evaluation of each component in a recognition pipeline. Our comparisons hold the trained classifier fixed to isolate the effect of calibration-label acquisition.

Uncertainty estimation is also important in HAR. Roy et al. study confidence-calibrated recognition [14], and hyperdimensional computing combined with conformal prediction supports uncertainty-aware inference on microcontrollers [15]. MiniRocket [16] and InceptionTime [17] provide established model families with contrasting representations. We use them to examine how acquisition behavior changes with the underlying score distribution.

2.2 Label-efficient stopping and its statistical target

Ilyas et al. [7] study a stream of n independent and identically distributed (i.i.d.) examples and the cost

$$C(m) = m + (n - m) \mathbb{E}[\|T_m(X)\|]/K. \quad (1)$$

Their procedure estimates normalized set size from candidate-label scores on unlabeled inputs and stops using a lower confidence bound on the cost increment. The regret analysis uses a monotone score-density-ratio assumption. The original-submission manuscript also describes a finite-sample stabilization for the experiments. These ideas motivate stopping decisions based on the observable size of prediction sets.

JRC targets containment of the prediction sets defined by a fixed reference pool. Its confidence statement is conditional on that pool and the uniform query order, with an additional size constraint on the monitoring batch. Table 1 summarizes the statistical targets of the related

approaches. The Ilyas comparison follows the original-submission version and distinguishes its theoretical stopping rule from the experimental stabilization.

Training-conditional validity has a longer history [18]. Liang et al. study conformal inference with early stopping of neural-network training and reuse of hold-out data [19]. Xu et al. develop active, anytime-valid risk-controlling sets using a labeling policy and risk-predictor assistance [20]. The August 2026 preprint by Skalse et al., *Online Conformal Prediction Beyond Feedback* (arXiv:2608.07139v1), studies a game with prediction and label-query actions on non-independent streams. Together, these studies address limited labeling resources through several population-risk and sequential-decision objectives.

2.3 Sequential finite-population inference

Finite-population sampling and hypergeometric inversion are standard statistical tools [21]. Time-uniform confidence sequences [8], sequential quantile estimation [9], sampling-without-replacement confidence sequences [10], and exact hypergeometric group-sequential tests [22] supply the closest foundations. The corrected conformal quantile and cumulative-probability scores are likewise established [4, 23]. JRC contributes a specialized joint rank-state calculation for a known finite horizon and reference rank. The experiments isolate the benefit of this calculation by applying the same reference pool, query order and set-inflation tolerance to established interval methods.

3 Finite-pool reference emulation

Figure 1 shows the frozen-model and query workflow. Algorithm 1 specifies the acquisition procedure.

3.1 Target, information and cost

Let a trained classifier be fixed before calibration acquisition. For a nonconformity score $s(x, y)$, its thresholded prediction set is

$$C(x; q) = \{y \in \mathcal{Y} : s(x, y) \leq q\}. \quad (2)$$

The label set $\mathcal{Y}$ is finite, with K classes. We fix a reference pool $\mathcal{P} = \{x_i\}_{i=1}^N$ before requesting its labels. The unknown true-label scores $S_i = s(x_i, y_i)$ are regarded as fixed values. Define

$$k = \lceil (N+1)(1-\alpha) \rceil, \quad q_{\text{ref}} = S_{(k)}, \quad (3)$$

with $q_{\text{ref}} = +\infty$ if $k > N$. The full-pool reference predictor is $C(x; q_{\text{ref}})$. Our target is to emulate this predictor with fewer queried labels. The reference rank uses the familiar split-conformal correction, but dependence within a pool does not make the resulting predictor marginally valid. Reference containment and population coverage are different statements.

We condition on all pool values, the trained classifier and a batch $\mathcal{X}_*$ of M unlabeled monitoring inputs. The probability statements concern a uniform permutation of the N pool indices, selected independently of their unrevealed labels. This design accommodates dependencies among people, repeated windows and trials because inference conditions on the complete finite pool.

All class scores on $\mathcal{X}_*$ are available without its labels. They determine when a quantile interval is sufficiently precise for the intended prediction sets. The size-control procedure is transductive: the monitoring inputs are available during acquisition, and the average inflation bound applies to that batch. Reference-set containment holds for every input on the confidence event.

The primary cost is the number T of queried window labels. When the pool involves m participants and each has setup cost c , a conservative secondary effort model is $T + cm$, against the reference cost $N + cm$. Every participant's setup is charged to both policies, even if stopping occurs before that participant's first queried window. These are modeled effort units. Presegmentation, obtaining sensor recordings and constructing the reference pool are outside the label-query savings.

3.2 Finite-population rank brackets

Choose checkpoints $\mathcal{T} \subseteq \{1, \dots, N\}$ with $N \in \mathcal{T}$. If scores tie, order the complete population by the pair (S_i, i) , using the fixed pool index to break ties. The numerical inequalities below remain valid when these pairs are projected onto their score values.

Table 1 Related statistical targets and acquisition decisions

Procedure	Sampling target	Stopping or acquisition decision	Guarantee considered
Ilyas et al. [7]	I.i.d. data-generating distribution	Cost-increment rule; empirical set size estimated from unlabeled inputs	Population coverage; asymptotic cost regret under a density-ratio condition
Howard and Ramdas [9]	Population quantiles from an i.i.d. stream	Any stopping time supported by the confidence sequence	Time-uniform confidence intervals for quantiles and distribution functions
Waudby-Smith and Ramdas [10]	Fixed population sampled without replacement	User-selected stopping applied to a confidence sequence	Time-uniform confidence sets for finite-population parameters
JRC	Corrected order statistic of a fixed finite reference pool	Endpoint set-size difference below a batch-specific tolerance	Reference-set containment for all inputs; average inflation control on the monitoring batch

The rows summarize the statistical targets. JRC evaluates acquisition rules under a common finite-reference containment requirement and monitoring-batch size tolerance

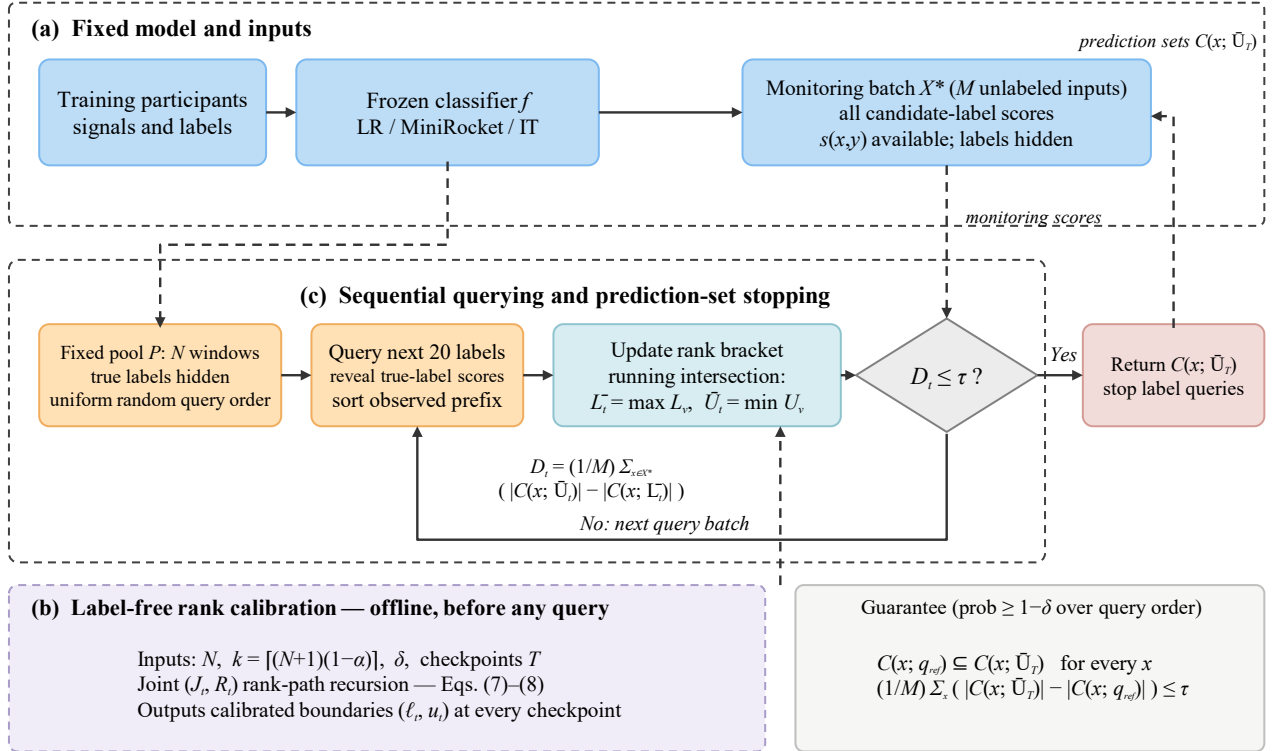


Fig. 1 Joint rank calibration workflow. Panel (a) defines the frozen classifier and available inputs; panel (b) constructs the rank boundaries before querying; panel (c) acquires labels and applies prediction-set stopping. Running maxima and minima use the visited checkpoints, and both sums range over the declared monitoring batch. The checkpoint set in panel (b) is $\mathcal{T}$; T denotes the stopping query count

Algorithm 1 Joint rank calibration for a fixed reference pool

Require: Frozen score function; fixed pool of N windows; uniform query permutation; monitoring scores; α, δ, τ ; checkpoints $\mathcal{T}$

- 1: Set $k = \lceil (N+1)(1-\alpha) \rceil$ and calibrate rank boundaries (ℓ_t, u_t) by the joint recursion
Joint recursion: Section 3.3, Eqs. (7)–(8).
 - 2: Initialize observed scores $\mathcal{S} \leftarrow \emptyset$, $L \leftarrow -\infty$, $U \leftarrow +\infty$
 - 3: **for** each checkpoint $t \in \mathcal{T}$ **do**
 - 4: Query only the new labels needed to reach t observations and add their true-label scores to $\mathcal{S}$
 - 5: Obtain the current rank bracket (L_t, U_t) from the ordered observed scores
 - 6: Form $(L', U') \leftarrow (\max\{L, L_t\}, \min\{U, U_t\})$
 - 7: **if** $L' > U'$ **then**
 - 8: Record an empty-intersection event and use $(L', U') \leftarrow (L_t, U_t)$
 - 9: **end if**
 - 10: Set $(L, U) \leftarrow (L', U')$ and calculate D_t from the monitoring scores using Eq. (9)
 - 11: **if** $D_t \leq \tau$ **then**
 - 12: **return** queried count t and predictor $C(x; U)$
 - 13: **end if**
 - 14: **end for**
-

Let J_t count the queried items whose full-population ranks are strictly below k , and let R_t indicate whether the item of rank k has appeared. Then $J_t + R_t$ counts sampled items with rank at most k . Their separate marginal distributions are

$$\begin{aligned} J_t &\sim \text{Hypergeom}(N, k-1, t), \\ J_t + R_t &\sim \text{Hypergeom}(N, k, t). \end{aligned} \quad (4)$$

Write $Q_p(N, a, t)$ for the smallest integer whose hypergeometric cumulative probability is at least p . At a common tail parameter η , use sample-rank boundaries

$$\ell_t = Q_\eta(N, k, t), \quad u_t = Q_{1-\eta}(N, k-1, t) + 1. \quad (5)$$

For the ordered observed scores $S_{(j)}^{[t]}$, set

$$L_t = S_{(\ell_t)}^{[t]}, \quad U_t = S_{(u_t)}^{[t]}. \quad (6)$$

Here ℓ_t is the lower sample-rank index defined in Eq. (5). The conventions are $S_{(0)}^{[t]} = -\infty$ and $S_{(t+1)}^{[t]} = +\infty$. At the census, both bounds are the known $S_{(k)}$. When $k > N$, both are $+\infty$ and the reference already contains every label.

On a path satisfying $J_t + R_t \geq \ell_t$ and $J_t < u_t$, at least ℓ_t observations have full rank at most k , while fewer than u_t observations have rank below k . Consequently, $L_t \leq q_{\text{ref}} \leq U_t$. A standard simultaneous-coverage construction

uses $\eta = \delta/[2(|\mathcal{T}| - 1)]$ and a union bound over the non-census checkpoints. The resulting Bonferroni hypergeometric (HG) brackets form the first baseline.

3.3 Joint calibration of the checkpoint boundaries

The union bound does not use the strong dependence between nearby checkpoints or between the two rank-count processes. We instead compute their joint crossing probability. Conditional on state $(J_t, R_t) = (j, r)$, the next observation has rank below k , equal to k , or above k , with probabilities

$$\frac{k-1-j}{N-t}, \quad \frac{1-r}{N-t}, \quad \frac{N-k-t+j+r}{N-t}, \quad (7)$$

respectively. They lead to states $(j+1, r)$, $(j, 1)$ and (j, r) . Impossible transitions have probability zero. Start with mass one at $(0, 0)$; propagate these probabilities and absorb mass at any checkpoint for which

$$j+r < \ell_t \quad \text{or} \quad j \geq u_t. \quad (8)$$

The absorbed mass $A(\eta)$ is the exact probability of at least one rank-boundary crossing under the finite query design. There are at most $2k$ states at each time. Direct propagation costs $O(Nk)$

operations and $O(k)$ working memory for a specified boundary family. It is performed before label queries and reused across pools with the same N , k , confidence level and checkpoints.

Increasing η weakens both tail protections, so $A(\eta)$ is nondecreasing, although discrete. We search this one-dimensional family by 24 bisection steps and retain a feasible value satisfying $A(\eta) \leq \delta - 10^{-10}$. The small numerical slack is implementation protection; the statistical claim concerns the exact recursion. Exact group-sequential calibration is established [22]; the contribution here is the coupled rank-state construction for preserving a finite calibration reference, together with the prediction-set stopping target.

3.4 Stopping in prediction-set space

Intersect valid past brackets by setting $\bar{L}_t = \max_{v \leq t} L_v$ and $\bar{U}_t = \min_{v \leq t} U_v$, where $v \in \mathcal{T}$. On the simultaneous coverage event this intersection contains q_{ref} . An empty intersection signals that event has failed; the implementation records it and falls back to the current bracket. At the census it returns the exact quantile regardless of previous events.

The stopping statistic is observable without monitoring labels:

$$D_t = \frac{1}{M} \sum_{x \in \mathcal{X}_*} [|C(x; \bar{U}_t)| - |C(x; \bar{L}_t)|]. \quad (9)$$

For a prespecified tolerance $\tau \geq 0$, stop at the first checkpoint with $D_t \leq \tau$ and return $C(x; \bar{U}_T)$. The census guarantees termination. The quantity τ is an absolute number of additional labels per input, not a normalized score width or coverage margin. An arbitrary numerical score tolerance would change meaning under a monotone reparameterization of the nonconformity score; this stopping statistic depends on actual prediction-set memberships.

Proposition 1 (reference preservation).

For a fixed classifier, finite reference pool, monitoring batch and declared checkpoints, suppose the joint recursion yields $A(\eta) \leq \delta$. Under uniform sampling without replacement, with probability at least $1 - \delta$, the returned predictor satisfies simultaneously

$$C(x; q_{\text{ref}}) \subseteq C(x; \bar{U}_T), \quad (10)$$

for every input x , and

$$\frac{1}{M} \sum_{x \in \mathcal{X}_*} [|C(x; \bar{U}_T)| - |C(x; q_{\text{ref}})|] \leq \tau. \quad (11)$$

No independence assumption on the fixed score values is required.

Proof. The recursion enumerates the law of the uniformly sampled rank path. Except on an event of probability $A(\eta)$, neither inequality in Eq. (8) holds at any checkpoint. The rank argument above gives $L_t \leq q_{\text{ref}} \leq U_t$ simultaneously. Running intersections therefore contain q_{ref} and are nonempty. Since the sets in Eq. (2) are nested in q , the upper endpoint gives Eq. (10) for every input. The reference set lies between the sets at the lower and upper endpoints, so its extra-label difference is bounded pointwise by the summand in Eq. (9). At the data-dependent stopping time, the average is at most τ . The confidence event covers every checkpoint and hence also this stopping time. $\square$

The containment event implies that empirical coverage cannot decrease relative to the same reference on any fixed labeled test collection. The confidence level is attached to one label-query campaign under the declared sampling design.

Strictly increasing score transformations preserve the rank ordering and every thresholded set membership, so they leave the query count and returned sets unchanged. This invariance provides an implementation check.

3.5 Implementation and comparison methods

Sorted monitoring scores allow Eq. (9) to be computed using two binary searches. Each checkpoint sorts the observed label scores. This direct implementation is adequate for the pool sizes studied; dynamic order-statistic structures could reduce repeated sorting for larger pools. Precomputed boundary tables are separate from per-pool query execution.

Both quantile-interval baselines satisfy the same finite-pool confidence constraint as JRC: conditional on the complete fixed pool, their intervals contain q_{ref} simultaneously at all declared checkpoints with probability at least $1 - \delta$ over uniform sampling without replacement. The comparison uses the same upper-endpoint output

and set-inflation stopping rule. Population coverage remains a separate property of the reference predictor.

The second quantile-interval baseline inverts the prior-posterior ratio confidence sequence (CS) of Waudby-Smith and Ramdas [10]. A uniform prior on the number of successes in the population induces the ordered-prefix mixture likelihood $B(j+1, t-j+1)$. For a hypothesized population count a , divide this mixture likelihood by

$$\mathcal{L}_t(a, j) = \frac{(a)_j (N-a)_{t-j}}{(N)_t}, \quad (12)$$

where $(a)_j$ denotes a falling factorial. Invert the resulting e-process separately at counts k and $k-1$, allotting each error $\delta/2$. The minimum accepted count at k gives the lower sample rank; one plus the maximum accepted count at $k-1$ gives the upper rank. We use the same running intersection, checkpoints and prediction-set stopping tolerance as the proposed engine.

We additionally compare ordinary conformal quantiles from fixed 25%, 50% and 75% prefixes, and an upper hypergeometric bound at a single fixed 50% look. The former have no reference-containment guarantee. The latter protects the reference at that fixed look but does not adapt label expenditure to a prescribed set-size tolerance. Diagnostic ablations omit time correction, omit running intersection, return a midpoint at the proposed stopping time, or use sparse checkpoints. Their different guarantees are kept explicit. The one-look HG comparator queries $\lceil N/2 \rceil$ labels and returns the upper endpoint of the same two-sided hypergeometric interval, with error 0.025 in each tail. Thus, “50% look” identifies the query fraction. This comparator uses a more conservative upper endpoint than a one-sided bound assigning error 0.05 to the upper tail.

4 Experimental design

4.1 Data and development history

The main evaluation includes six development datasets: UCI HAR [24], DSADS [25], MHEALTH [26], WISDM [27], PAMAP2 [28] and REALDISP [29]. They contain 122 eligible participant identifiers and 54,482 retained windows. These

datasets were examined during acquisition-policy development. Earlier risk-based and randomized-allocation rules showed poor tolerance transfer on PAMAP2 and REALDISP, motivating the fixed-reference target. Their protocols and aggregate negative results are retained in Online Resource 1.

USC-HAD [30] provides the main confirmation benchmark. The method, confidence level, primary tolerance, query schedule, preprocessing and splits were locally locked on 12 September 2026 before model predictions were available. All 14 participants meet the requirement of at least 120 retained windows, yielding 5,353 windows and 12 classes. The main seven-dataset study therefore contains 136 dataset-specific participant identifiers and 59,835 windows.

The selected input is three acceleration axes resampled to 128 time points. UCI HAR uses the release’s 2.56-second total-acceleration windows, retaining alternating windows per subject; its release ordering is not verified raw chronology. DSADS uses five-second torso-acceleration segments. MHEALTH uses five-second chest windows; WISDM uses five-second smartphone-acceleration windows from its 18-activity release. PAMAP2 uses protocol chest acceleration, and REALDISP uses ideal-placement BACK acceleration. The original REALDISP manual places BACK third in the log, unlike the fifth position listed in the repository webpage summary; we follow the original manual. The retained sample counts and preprocessing provenance are included in Table 2 and the supplement. Different window durations in the original HAR release prevent interpreting the seven datasets as identical sensor protocols.

For USC-HAD, each original MAT trial remains separate. We take nonoverlapping 500-sample windows at the documented 100 Hz, retain acceleration columns 1–3, reject nonfinite windows and incomplete tails, and use anti-aliased polyphase resampling to 128 points. Dataset-provided trial segmentation is treated as available; our label-cost model does not charge or solve raw-stream segmentation. The filename convention supplies activity IDs. An independent schema audit finds one conflict: Subject14/a3t2.mat is activity 3 by filename but activity 2 (walking-left) in its internal metadata. Main labels follow the locked filename rule. A separate sensitivity excludes that trial, refitting classifiers whenever it

occurs in training. Another file lacks the optional numeric activity field but has a matching activity-name string; it is retained. No exact duplicate signal arrays are found among the 840 trials. The main analysis and the exclusion sensitivity are both reported.

4.2 Frozen classifiers and subject partitions

Every acquisition method uses identical probability vectors within a classifier case. Logistic regression (LR) uses 35 standardized statistical features, regularization $C = 1$ and a maximum of 2,000 iterations. MiniRocket (MR) uses 2,016 kernels followed by a standardized logistic-regression head. The InceptionTime-style model (IT) uses six modules, 32 filters per branch, kernel lengths 39, 19 and 9, residual connections every third module, global pooling and a classification head. It is trained for 40 epochs with Adam, learning rate 10^{-3} and batches of 128, without a test-selected checkpoint. The statistical features, normalization and model code are shared with the preceding frozen model study. Scikit-learn and SciPy support the classical pipeline [31, 32].

We reuse 249 cases from 83 outer partitions on the six development datasets. Each case has disjoint training, calibration and test participants. The earlier study trained auxiliary out-of-fold models, but this method does not require their predictions or their fit cost. USC-HAD has five fixed random seeds and seven test folds per seed: two test participants, eight training participants and four calibration participants in each fold. This gives 35 partitions and 105 new final-classifier fits. The 354 model cases reuse the 83 development and 35 confirmation partitions; repeated results are aggregated within participants.

The least ambiguous set-valued classifier (LAC) score [33] is $s(x, y) = 1 - \hat{p}_y(x)$. The secondary score uses adaptive prediction sets (APS) [23] in a deterministic inclusive form: sort class probabilities in descending order, with stable class-index ties, and assign each class its cumulative probability. There is no randomization within the APS score and no forced top-label insertion. Empty and full sets remain in the calculations. The same scoring convention is used for the reference and every acquisition comparator, and $\alpha = 0.1$ throughout.

4.3 Reference pools and query experiments

We construct each reference pool from m calibration participants. The requested ceilings are $B \in \{120, 240, 480, 960\}$, and the actual pool size is $N = \min(B, 120m)$. Participant quotas differ by at most one; a random participant ordering allocates any remainder. Each quota is sampled uniformly without using its labels, after which the complete selected pool is independently permuted. When several ceilings produce the same pool size, the evaluation retains the declared configurations and reports their actual N .

The primary settings are confidence error $\delta = 0.05$, mean set-inflation tolerance $\tau = 0.5$, and checkpoints every 20 queried labels plus the census. The tolerance is measured in additional candidate labels per monitoring input. Secondary tolerances are 0.1, 0.25 and 1.0. Forty paired randomizations of pool selection and query order are used for each classifier case and ceiling. Every comparator shares the seeds, participants and windows. Complete labels define the evaluation reference; an oracle-based replay checks that acquisition decisions use only the labels already queried.

The complete test-input batch supplies monitoring scores with true labels hidden. The stopping statistic averages set size over windows in that batch. Empirical coverage and set size are also summarized with equal weight per participant. These two aggregations are retained explicitly in the result tables.

4.4 Metrics, uncertainty and ablations

Primary outcomes are queried-label savings $1 - T/N$, reference-set containment failure and batch size inflation. A containment failure is the conservative scalar event $\bar{U}_T < q_{\text{ref}}$; with ties or gaps in candidate scores, this can flag a threshold failure even when no observed test set loses a label. Bracket failure checks both endpoints. Size failure checks whether actual mean extra labels on the monitoring batch exceed τ . Each query campaign is counted once, regardless of the number of held-out people.

For a test participant with empirical coverage c_u , positive shortfall is $(0.9 - c_u)_+$. The

Table 2 Public datasets and retained samples

Dataset	People	Windows	Classes	Selected signal	Seconds
UCI HAR	30	5,155	6	Waist total acceleration	2.56
DSADS	8	9,120	19	Torso acceleration	5
MHEALTH	10	1,335	12	Chest acceleration	5
WISDM	51	32,653	18	Phone acceleration	5
PAMAP2	8	3,775	12	Chest acceleration	5
REALDISP	15	2,444	33	BACK acceleration	5
USC-HAD	14	5,353	12	Front right hip acceleration	5

The first six datasets are development and reanalysis; USC-HAD is the locked confirmation benchmark. All signals have three axes and are resampled to 128 points. Eligibility requires at least 120 windows per participant. Sample counts refer to the fixed main preprocessing, before the separate USC-HAD schema sensitivity.

positive part is calculated before averaging randomizations. Accuracy and macro-F1 describe the frozen point classifiers. Subject summaries average paired randomizations and repeated partitions; paired 95% bootstrap intervals resample these summaries within each dataset. The intervals therefore describe participant variation conditional on the fitted models and partitions.

JRC, Bonferroni HG and the confidence sequence based on the prior-posterior ratio use identical tolerances and query checkpoints. Ablations remove the running intersection, use a sparse checkpoint grid, omit repeated-look correction or return the midpoint at the proposed stopping time. The uncorrected and midpoint variants serve as diagnostic controls. Fixed fractions and a one-look HG rule provide additional label-cost and set-size trade-offs. Exploratory checks vary δ and compare monitoring on even-indexed inputs with evaluation on odd-indexed inputs in the first MR case of each development dataset.

All runs use an Intel Core i9-12900HX CPU and an RTX 4070 Laptop GPU with 8 GB memory for neural training. We measure boundary-table construction separately from online acquisition calculations. Human annotation duration and device energy were outside the computational benchmark. Exhaustive small-population enumeration checks both rank engines and the coupled recursion; additional checks cover ties, census behavior, oracle access and strictly monotone score transformations.

4.5 Additional HHAR confirmation

After the main seven-dataset study, an additional Heterogeneity Human Activity Recognition

(HHAR) confirmation [34] was specified and publicly committed on 14 September 2026 before preprocessing outcomes or model fitting. The frozen method and classifier implementations were retained. All smartphone-accelerometer streams were processed separately by user and device. Five-second nonoverlapping windows were formed within activity-consistent runs, splitting at non-increasing timestamps or gaps above 0.5 s. Windows required at least 60 finite samples and 4.9 s of observed duration. Timestamp interpolation at 200 Hz followed by anti-aliased resampling produced 128 points. A label-blind random cap of 1,200 windows per user retained 10,800 windows from nine users and six classes. The fixed continuity filter retained very few Nexus4 windows. Device-specific retention counts are provided in Online Resource 1.

Five declared seeds and three user-disjoint test folds used four training, two calibration and three test users per fold, yielding 45 additional LR, MR and IT cases. The primary analysis used LAC, $N = 240$, $\delta = 0.05$, $\tau = 0.5$ and 40 paired query orders; $N = 120$ and APS were secondary. Alongside Bonferroni HG, the fixed Beta(9,1) confidence-sequence prior was a primary comparator, chosen from the preceding diagnostic before HHAR outcomes. User-level paired differences and 95% bootstrap intervals follow the declared aggregation. This cohort is reported separately from the main USC-HAD confirmation.

5 Results

5.1 Label savings against sequential quantile baselines

Table 3 reports absolute reference utility and label savings for the three interval methods. Figure 2 adds the secondary APS score. All engines use the same confidence error, tolerance, pool and query order. This common setting isolates the acquisition efficiency of their quantile bounds.

On USC-HAD, the joint engine saves 23.67% of reference labels, compared with 17.91% for HG and 11.74% for the uniform-prior CS. Its mean coverage is 91.80%, against 90.36% for the reference, and mean set size changes from 2.590 to 2.769. The coverage increase comes from the more inclusive calibration threshold; the point classifiers are unchanged.

The six development datasets show a substantial range. Joint-engine savings are 71.62% on HAR and 45.17% on MHEALTH, but only 4.19% on DSADS and 6.59% on WISDM. PAMAP2 and REALDISP lie between these extremes. Large absolute savings are therefore not a universal property of certified stopping. Broad reference sets and score distributions with many decision changes near the reference quantile can make the tolerance costly to certify.

5.2 Reference preservation and absolute utility

Table 4 and Fig. 3 compare fixed-fraction reductions with the quantile-interval baselines. Fixed fractions provide substantial label savings by construction, but their ordinary prefix quantiles need not contain the full-reference sets. A midpoint at a certified stopping time similarly discards the reason that upper-endpoint containment holds. The one-look HG rule returns the upper endpoint of a two-sided interval at the predetermined 50% query fraction; it does not impose the adaptive set-inflation tolerance.

The proposed confirmation threshold-failure rate is 2.04%, and its batch size-failure rate is 0.72%. The confidence statement applies to one query campaign. A failure of the scalar threshold bracket may not remove any label on the finite observed input batch if no candidate score lies in the gap.

Reference quality explains important differences across datasets. Some development references fall below nominal 90% coverage, and WISDM returns broad candidate sets. These cases show where improving the recognition or calibration stage remains important.

5.3 Paired improvements and classifier dependence

Figure 4 reports paired participant bootstrap intervals for additional savings against the quantile-interval baselines. These intervals condition on the fitted models and split collection. Table 5 separates point-classification performance from acquisition outcomes for each confirmation backbone. The acquisition policy does not retrain or select among these classifiers.

5.4 Tolerance, query checkpoints and mechanism ablations

Figure 5 reports all four declared tolerances. Relaxing the allowed average number of additional labels permits earlier stopping, as expected from Eq. (9). The tolerance controls the permitted added ambiguity and the resulting cost-utility trade-off. The primary tolerance remains 0.5 on confirmation; it was not selected by maximizing the confirmation savings.

Table 6 separates the mechanism choices. Removing running intersections reduces confirmation savings from 23.67% to 21.10%; sparse checkpoints reduce them to 16.84%. Returning the midpoint retains the same label count but raises threshold failure to 46.40%, illustrating why the protected upper endpoint is essential. Omitting repeated-look correction increases savings to 33.71% and threshold failure to 6.29%, above the declared 5% error level.

Figure 6 traces the first declared USC-HAD MR query campaign at a 480-window pool. This case is selected by its fixed index, not by searching for the largest gain. The census collapses all valid constructions to the same full-reference quantile. Before that point, the mapping from rank uncertainty to set membership determines when each engine meets the tolerance.

Figure 7 relates label savings to absolute reference set size and shows the corresponding reference and returned-set coverage pairs.

Table 3 Primary LAC results at confidence error 0.05 and tolerance 0.5

Dataset	Ref. cov. (%)	Ref. size	Joint saved (%)	HG saved (%)	CS saved (%)	Joint extra size
UCI HAR	90.45	1.035	71.62	64.93	58.97	0.235
DSADS	88.69	5.015	4.19	2.71	1.10	0.073
MHEALTH	89.07	1.459	45.17	38.25	30.92	0.206
WISDM	89.92	8.996	6.59	4.35	2.49	0.122
PAMAP2	87.14	2.933	16.20	12.02	6.77	0.151
REALDISP	89.80	2.377	18.56	13.77	8.82	0.163
USC-HAD	90.36	2.590	23.67	17.91	11.74	0.179

Saved (%) = $100(1 - T/N)$: window labels saved relative to the full reference pool of N labels, excluding training, segmentation and setup costs. HG: Bonferroni hypergeometric brackets. CS: prior-posterior ratio rank confidence sequence. Coverage and set size give equal weight to participant summaries. Extra size is relative to the full reference. The stopping guarantee concerns the window-weighted monitoring batch.

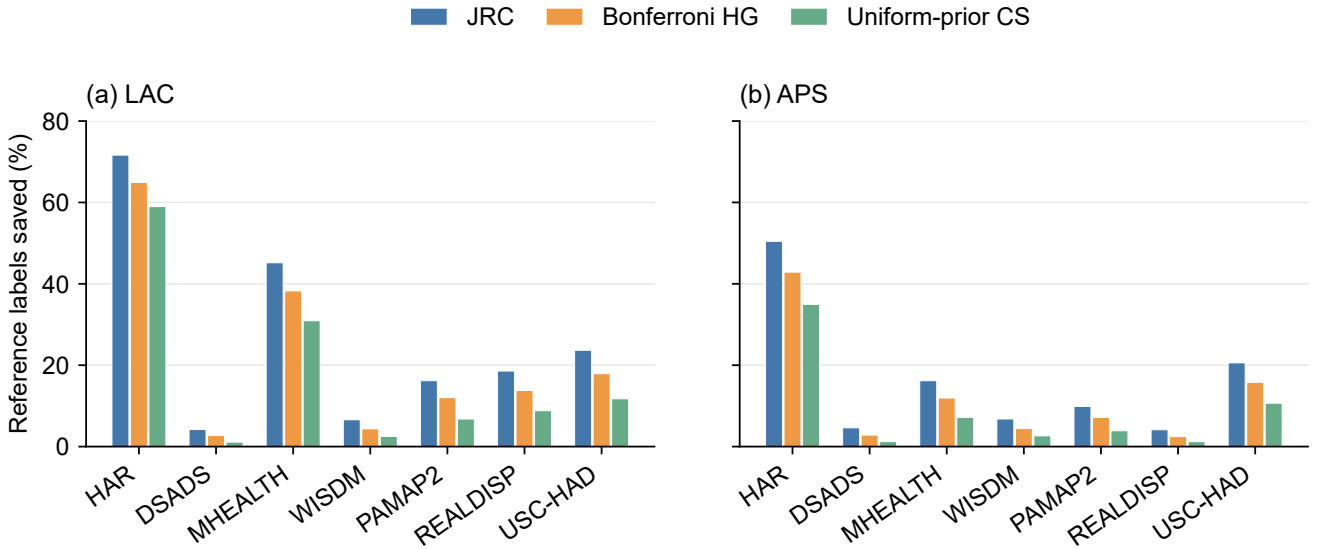


Fig. 2 Main-study reference-label savings: (a) LAC; (b) APS, at confidence error 0.05 and tolerance 0.5. Within each dataset, bars are ordered JRC, Bonferroni HG and the primary uniform-prior confidence sequence from left to right. Both panels use the same zero-based scale. Values average the declared pool ceilings and participant summaries; paired uncertainty is reported in Section 5.3

5.5 Sensitivity to stronger confidence-sequence priors

The primary confidence-sequence baseline uses a uniform prior on the finite-population success count. After the main confirmation, we evaluated four fixed beta-binomial priors across all 354 existing classifier cases. A $\text{Beta}(a, b)$ mixture replaces the ordered-prefix mixture likelihood by $B(j+a, t-j+b)/B(a, b)$ while retaining the same likelihood-ratio construction and error allocation. Exact rank-path checks verify the error bound for each finite design used. The diagnostic reuses the

main models and queries after confirmation; the locked JRC rule is unchanged.

Table 7 reports all four prior choices. Tail-centered priors improve the baseline, showing that the uniform-prior comparison alone understates the performance of this method family. On USC-HAD, the strongest of the tested alternatives saves 15.33% of labels, compared with 11.74% for the primary uniform prior and 23.67% for JRC. JRC retains the largest dataset-average savings among these choices on all seven datasets. The comparison covers the four specified priors and the declared benchmark designs. Reconstructing

Table 4 USC-HAD primary results and run-level failures

Policy	Saved (%)	Coverage (%)	Set size	Window extra	Threshold fail (%)	Size fail (%)
Full reference	0.00	90.36	2.590	0.0000	0.00	0.00
Joint rank engine	23.67	91.80	2.769	0.1824	2.04	0.72
Bonferroni HG	17.91	91.86	2.774	0.1870	0.45	0.21
Uniform-prior CS	11.74	91.76	2.761	0.1735	0.02	0.05
Fixed 25%	75.00	90.46	2.713	0.1220	42.26	15.03
Fixed 50%	50.00	90.41	2.632	0.0416	41.42	6.09
Fixed 75%	25.00	90.36	2.601	0.0102	39.83	2.20
One-look HG (50% look)	50.00	93.83	3.356	0.7737	1.73	46.64
Midpoint output	23.67	90.04	2.562	-0.0282	46.40	0.14
No time correction	33.71	91.73	2.766	0.1784	6.29	3.02

Saved (%) = $100(1 - T/N)$: window labels saved relative to the full reference pool of N labels, excluding training, segmentation and setup costs. Coverage and set size use participant weighting. Window extra is mean set inflation on the monitoring windows, as in Eq. (11), averaged equally over the declared query campaigns. One-look HG uses 2.5% per tail and returns only the upper endpoint. Threshold failure is upper output threshold below the complete-reference threshold. Size failure is monitoring-batch inflation above 0.5 labels. Rates average paired query campaigns, each counted once. Fixed fractions, midpoint output and the uncorrected rule do not have the proposed joint contract.

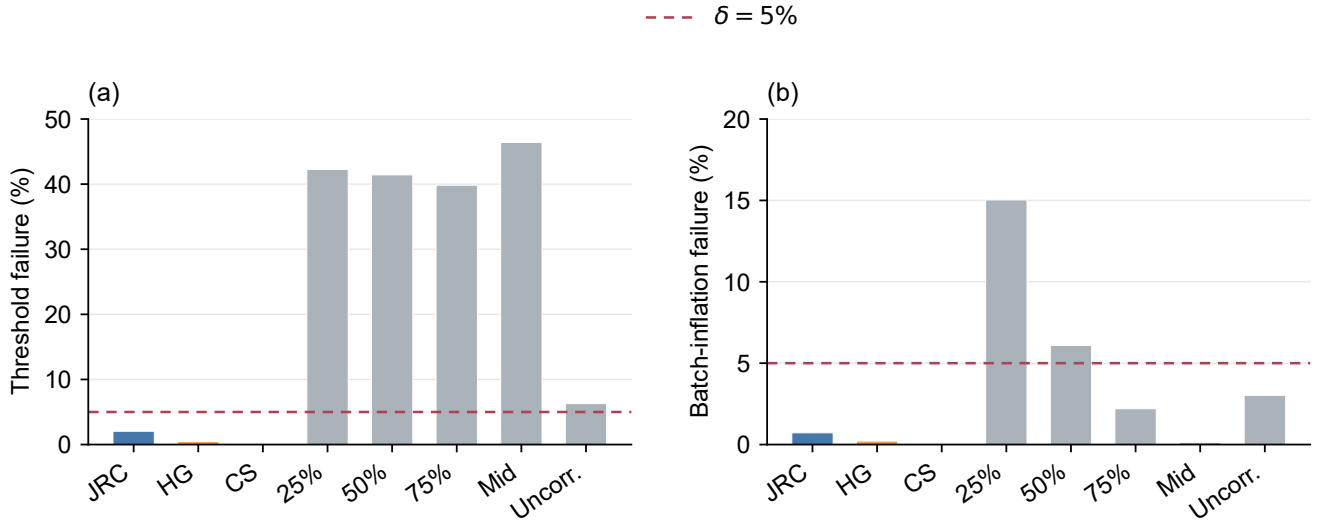


Fig. 3 USC-HAD LAC failure diagnostics: (a) output threshold below the full-reference threshold; (b) monitoring-batch inflation above 0.5 labels. The dashed line is $\delta = 5\%$. The first three bars are the interval methods; gray bars are diagnostic rules without the same guarantee. The labels 25%, 50% and 75% denote fractions of reference labels queried; Mid is midpoint output and Uncorr. is uncorrected repeated looks. Numerical values are reported in Table 4

Table 5 USC-HAD point-classifier and acquisition results

Classifier	Accuracy (%)	Macro-F1 (%)	Labels saved (%)	Set coverage (%)	Set size
LR	60.17	56.14	32.85	92.76	2.769
MR	68.04	65.40	23.31	92.82	2.573
IT	65.14	60.32	14.85	89.83	2.967

Each classifier has 35 main outer partitions. IT denotes a single InceptionTime-style network. Macro-F1 includes all 12 classes, with zero contribution for undefined class scores according to the supplied implementation.

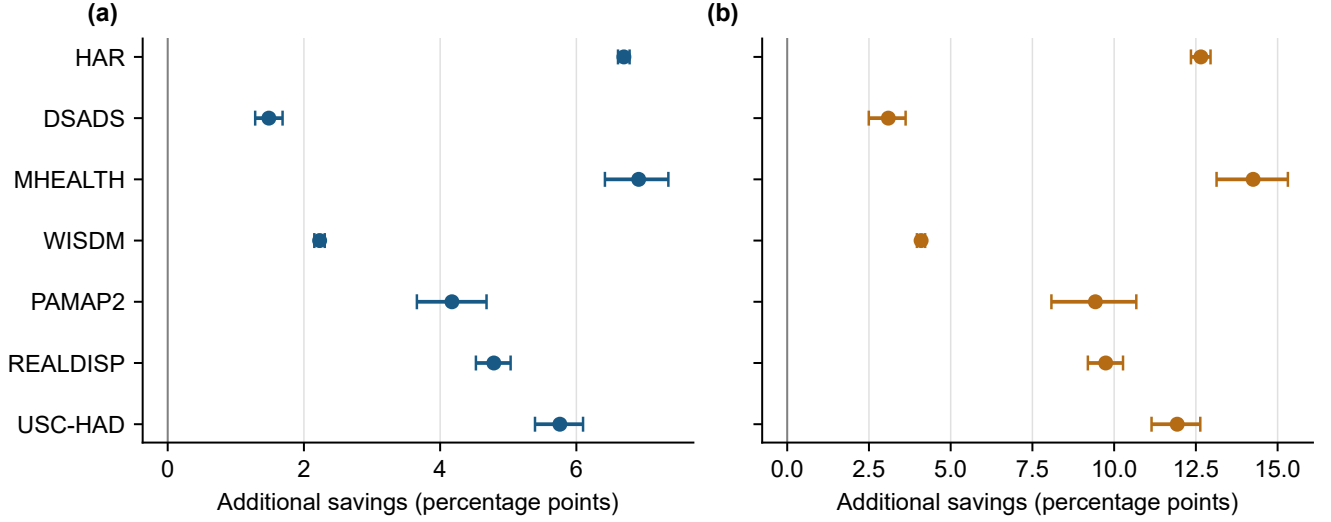


Fig. 4 Additional LAC label savings of JRC relative to (a) Bonferroni hypergeometric brackets and (b) the primary uniform-prior confidence sequence. Points are paired participant means; bars are 95% bootstrap intervals conditional on the fitted models and partitions

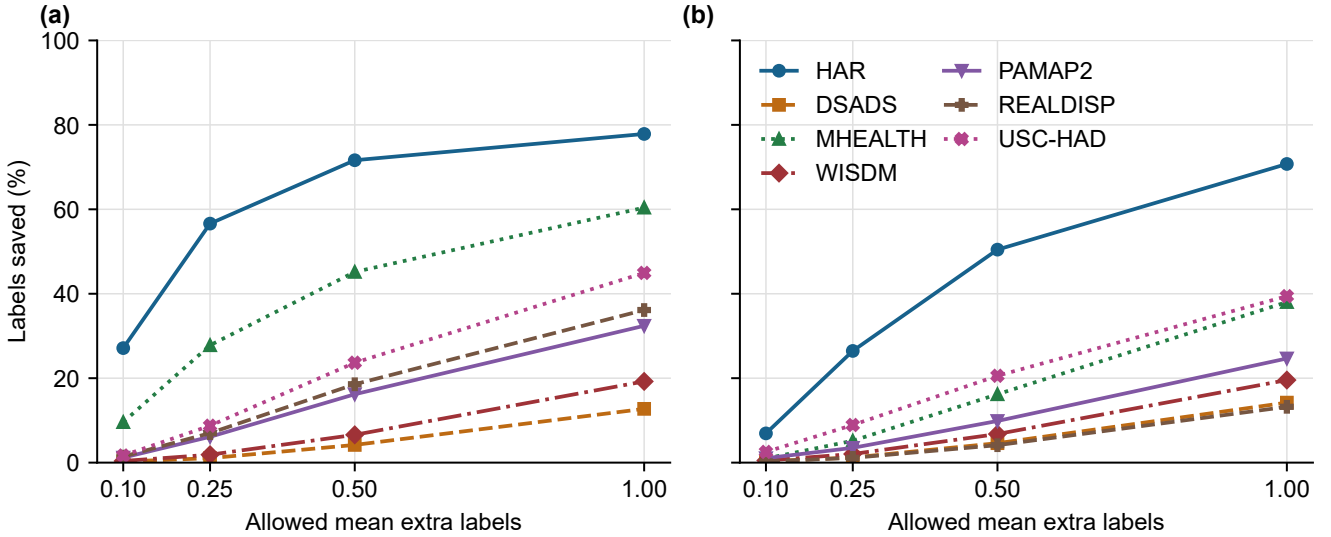


Fig. 5 Tolerance-savings curves for JRC: (a) LAC; (b) APS. Distinct markers and line styles identify the datasets. All four tolerances were declared before USC-HAD results. Values are means; primary-tolerance paired uncertainty is reported in Fig. 4

JRC and the uniform-prior results reproduces the frozen dataset averages within 10^{-12} .

5.6 Schema exclusion and computation

The exclusion sensitivity removes the USC-HAD trial with conflicting activity identifiers from every data role. The original classifier is reused only when that trial was absent from its training set;

Table 6 USC-HAD mechanism ablations

Rule	Saved (%)	Extra size	Threshold fail (%)	Size fail (%)
Joint rank engine	23.67	0.179	2.04	0.72
No intersection	21.10	0.178	0.67	0.06
Sparse checkpoints	16.84	0.079	1.77	0.36
Midpoint output	23.67	-0.028	46.40	0.14
No time correction	33.71	0.175	6.29	3.02

Saved (%) = $100(1 - T/N)$: window labels saved relative to the full reference pool of N labels, excluding training, segmentation and setup costs. Only the first three rows retain the declared guarantee with their respective boundaries and checkpoints. The midpoint has the same stopping labels as the joint engine but a different output rule. The uncorrected rule reuses fixed-time intervals at multiple looks.

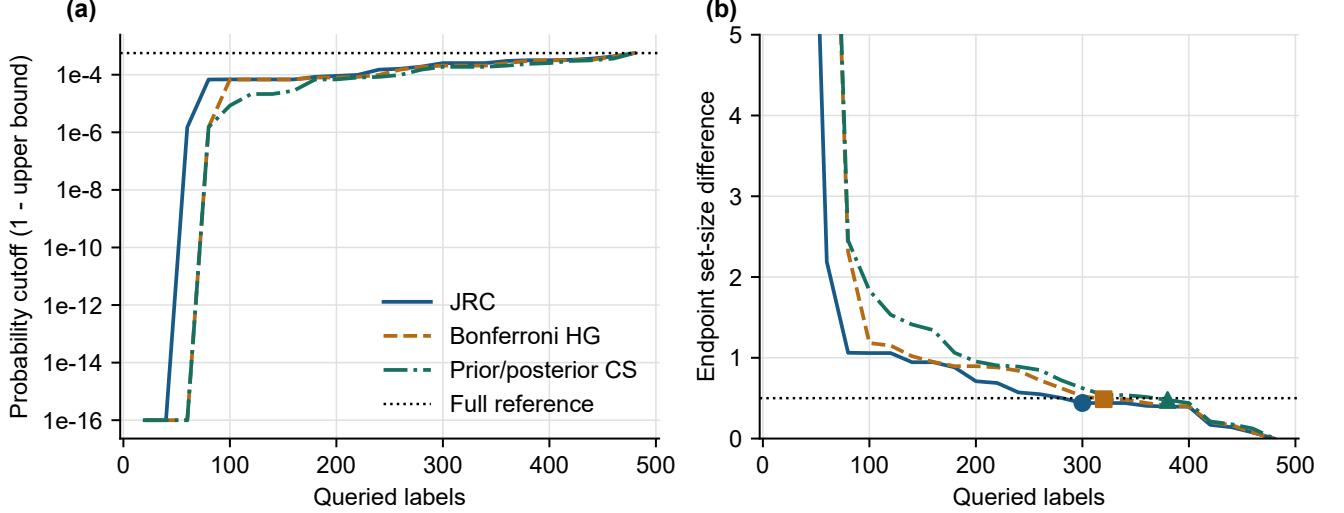


Fig. 6 First declared USC-HAD MR query run, LAC, 480 reference windows: (a) running upper quantile endpoints represented as the corresponding probability cutoff $1 - \bar{U}_t$ on a logarithmic axis; (b) endpoint set-size difference with stopping locations marked. The reference quantile is approximately 0.999436. Zero cutoffs and infinite upper bounds are displayed at 10^{-16} in (a); this is only a display convention. The horizontal line in (b) is the 0.5-label tolerance

otherwise it is refitted. Joint-engine label savings change from 23.67% to 22.70%, coverage from 91.80% to 92.30%, and mean set size from 2.769 to 2.814. This check measures sensitivity to the ambiguity; it does not establish which of the conflicting original labels is correct. The locked main data and outcomes remain intact.

Boundary tables depend only on the finite design and can be reused. Table 8 separates table construction from execution with a cached table. Timings exclude JIT compilation, model inference and human labeling. The online benchmark uses a fixed synthetic score array and 1,000 unlabeled inputs with 12 candidate classes to isolate computation; this benchmark isolates computation from recognition quality.

5.7 Additional HHAR confirmation

Table 9 reports the additional primary confirmation. JRC saves 34.03% of reference labels, compared with 28.58% for HG and 25.13% for the pre-specified Beta(9,1) CS. The paired differences are 5.45 percentage points (95% CI 5.09 to 5.85) against HG and 8.89 percentage points (95% CI 8.52 to 9.23) against the stronger-prior CS. These intervals describe participant variation conditional on the fitted models and partitions.

The full reference has mean coverage 88.10% and mean set size 1.807; JRC returns 90.84% coverage and mean set size 1.997. Threshold and monitoring-batch inflation failures are reported alongside savings. Secondary pool-size, APS and classifier-specific results are retained in Online

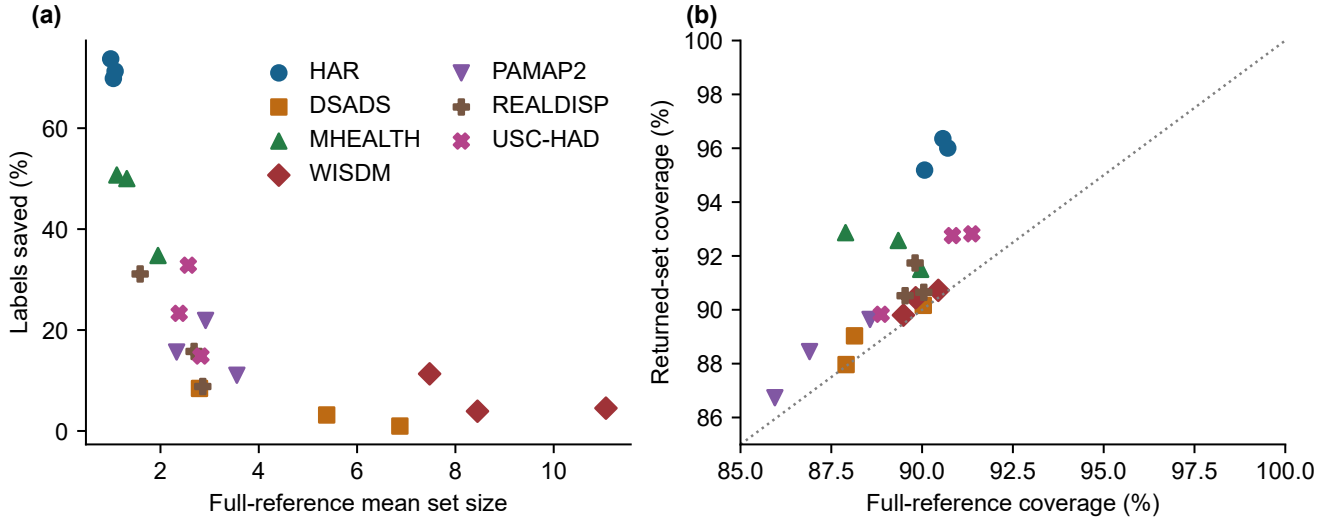


Fig. 7 Reference utility and acquisition behavior for LAC: (a) label savings versus full-reference set size; (b) returned-set versus reference coverage. Each point represents one dataset-classifier mean; dataset identity is encoded by both marker shape and color. Each dataset contributes LR, MR and IT points, whose identities and values are supplied in the model-summary table of Online Resource 1. The diagonal indicates equal coverage

Resource 1. This additional cohort supports the acquisition gain on a separately collected dataset under the fixed continuity rules.

Table 7 Post-confirmation prior sensitivity: percentage of reference labels saved

Dataset	JRC	CS(1, 1)	CS(0.5, 0.5)	CS(1.8, 0.2)	CS(9, 1)
UCI HAR	71.62	58.97	60.85	66.89	64.84
DSADS	4.19	1.10	1.12	2.09	1.82
MHEALTH	45.17	30.92	32.06	36.58	36.51
WISDM	6.59	2.49	2.53	3.02	3.34
PAMAP2	16.20	6.77	6.95	9.24	9.20
REALDISP	18.56	8.82	9.16	10.86	11.73
USC-HAD	23.67	11.74	12.01	14.22	15.33

LAC, confidence error 0.05, tolerance 0.5 and the original paired query orders. $CS(a, b)$ uses a $Beta(a, b)$ mixing distribution. Values give equal weight to participant summaries. All four choices are reported; this diagnostic follows the main confirmation

Table 8 Boundary construction and cached acquisition computation

Pool size	Build table (ms)	Cached query loop (ms)	Exact crossing probability
120	8.31	0.18	0.04843
240	25.07	0.26	0.04606
480	53.80	0.50	0.04991
960	153.41	0.45	0.04981

Five repeated timings per size on the stated CPU. All confidence errors are 0.05. Boundary construction is a one-time cost for each distinct pool and checkpoint design; query execution includes sorting monitoring scores and observed prefixes.

Table 9 Additional HHAR primary confirmation, $N = 240$

Policy	Saved (%)	Threshold fail (%)	Inflation fail (%)
Reference	0.00	0.00	0.00
JRC	34.03	1.33	0.28
HG	28.58	0.39	0.00
CS(9, 1)	25.13	0.28	0.00
CS(1, 1)	20.64	0.06	0.00

Saved (%) = $100(1 - T/N)$ relative to the complete reference pool, excluding training, segmentation and setup costs. $CS(a, b)$ uses a $Beta(a, b)$ prior; CS(1, 1) is uniform. Failure rates count query campaigns once.

6 Discussion and limitations

JRC improves label efficiency by calibrating the joint rank path across checkpoints. The benefit is greatest when the reference quantile can be localized while adding few candidate labels. The main experiments and the separately specified HHAR confirmation support this mechanism. Stronger confidence-sequence priors recover part of the gap to the uniform-prior baseline, while JRC retains an advantage in the reported dataset averages.

The reference predictor determines the practical value of this preservation guarantee. Broad prediction sets or weak reference coverage call for improvements to recognition and calibration. The containment statement applies to every input, while the size-inflation bound concerns the specified monitoring batch. The confidence-level checks and the comparison of even- and odd-indexed monitoring inputs in Online Resource 1 characterize sensitivity to these choices.

The evaluation combines six development datasets with the main USC-HAD confirmation and a separately specified HHAR confirmation. The fixed HHAR continuity rule leaves very few Nexus4 windows, so its evidence concerns the retained records. Further deployment conditions and practical annotation studies would extend this evidence. In particular, scattered window queries can incur context-switching costs: the modeled effort $T + cm$ captures participant setup but omits human annotation duration and segmentation effort. Extending the sampling law to clustered or nonuniform queries would support those workflows.

7 Conclusion

Joint rank calibration reduces the labels required to preserve a fixed calibration reference. Its coupled recursion controls quantile-boundary crossings across the declared checkpoints, and its stopping rule measures the resulting prediction-set ambiguity on unlabeled inputs. The main evaluation and the additional HHAR confirmation show useful savings over the tested interval comparators. The results also identify low-saving cases and the importance of reference quality, providing a concrete basis for choosing when to apply the method.

Data availability

The eight public datasets analyzed in this study are available from their original sources [24–30, 34]. Frozen classifier predictions, evaluation tables and protocol records supporting the reported results are provided in Online Resource 1.

Code availability

The acquisition algorithms, analysis code and reproduction instructions are available at <https://github.com/Cyrano666/jrc-reference-emulation>. Online Resource 1 contains the accompanying code snapshot.

Supplementary information

Online Resource 1: reproduction code, frozen predictions, evaluation tables, protocol records, verification results and editable figure sources.

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